

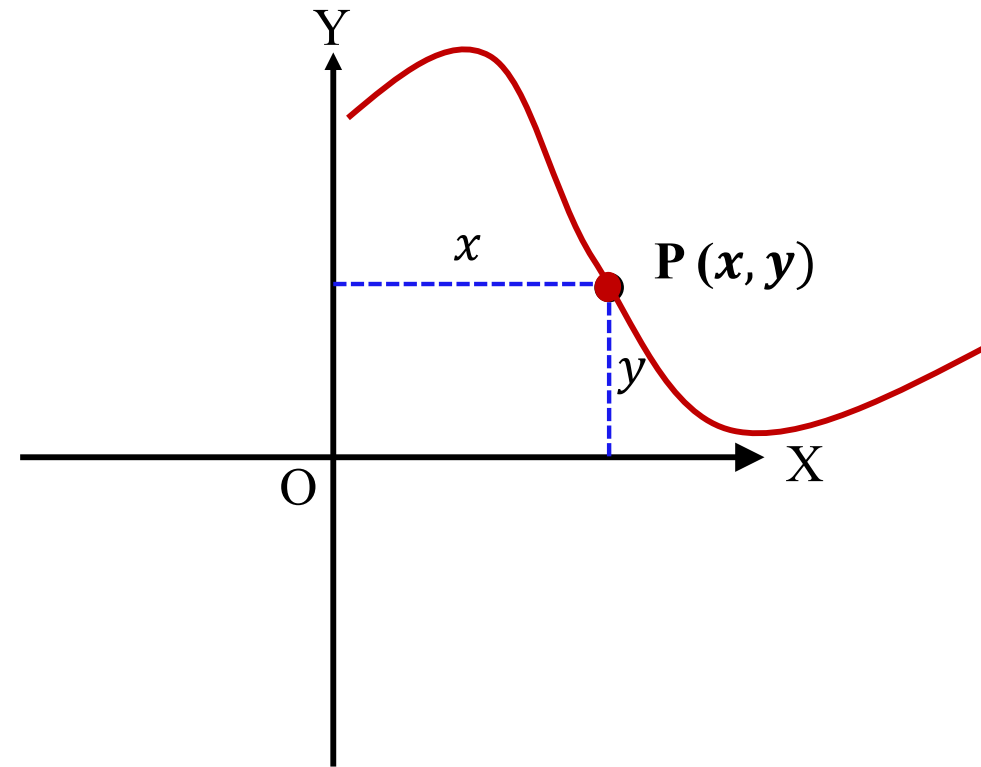
## Lecture 03

# Coordinate systems



# Cartesian coordinate system in plane

- Instantaneous position  $P$  can be represented by measuring its perpendicular distances ( $x$  and  $y$ ) from two perpendicularly intersecting lines (called Axes) passing through the origin.
- In Cartesian coordinate, the position  $P$  is represented by  $(x, y)$ .

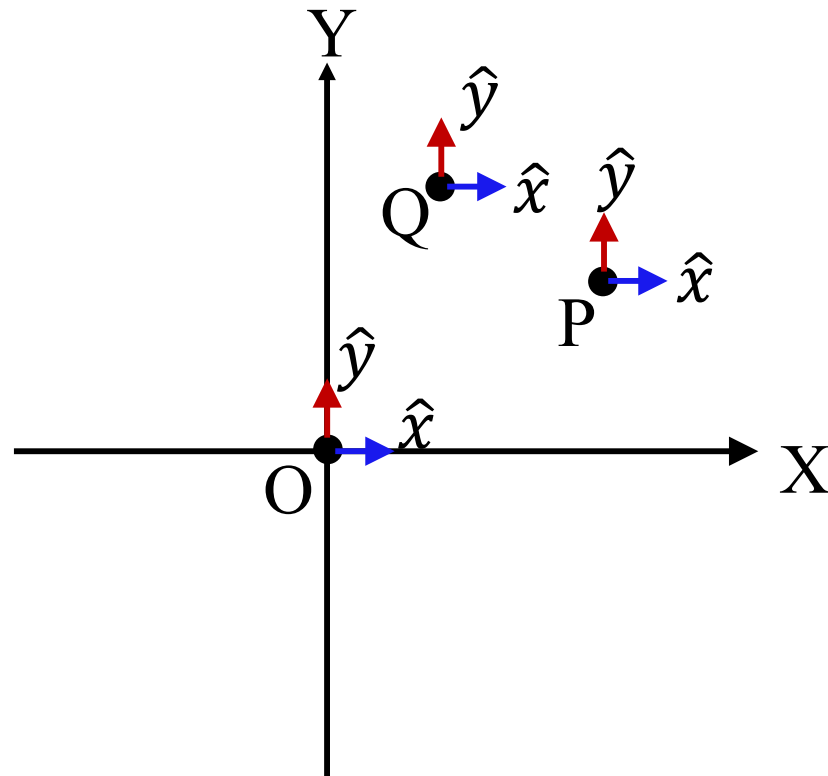


Cartesian coordinate system in plane

All points in the plane are therefore referred with respect to same set of X and Y.



# Unit vectors in the plane: Cartesian



- In Cartesian coordinate system, one can define two unit vectors  $\hat{x}$  and  $\hat{y}$  at every point in the plane.
- $\hat{x}$  and  $\hat{y}$  are unit vectors in the increasing direction of  $x$  and  $y$ , thus they are parallel to coordinate axes  $X$  and  $Y$ , respectively.
- $\hat{x}$  and  $\hat{y}$  are orthogonal, and directed in the same direction at every points.

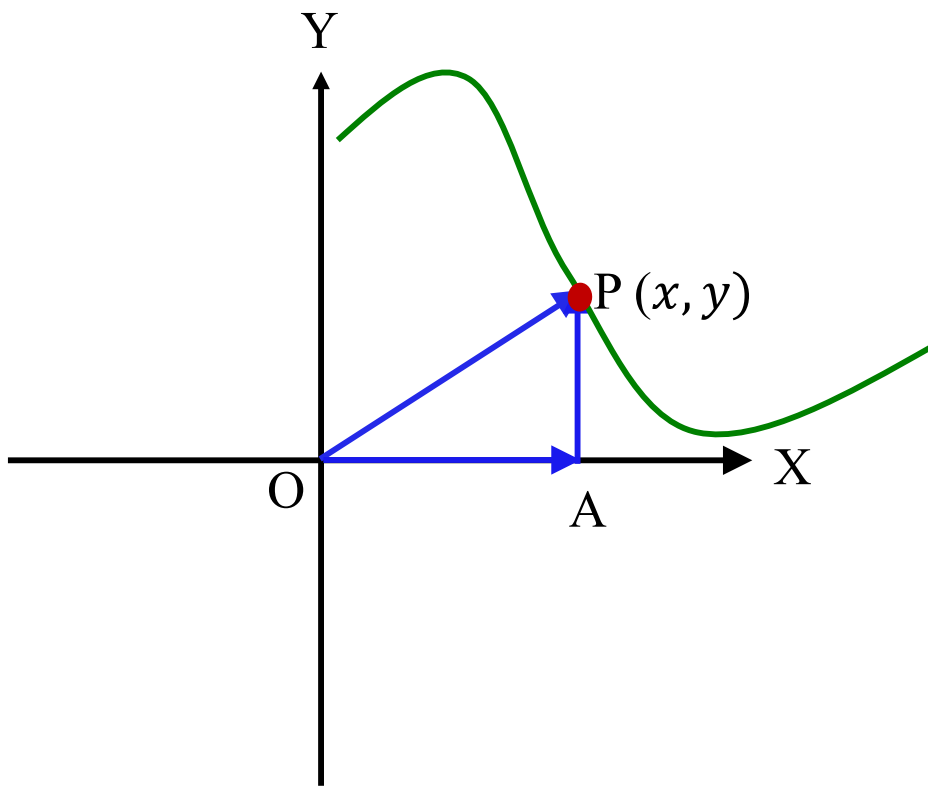
## Another way of looking unit vector Cartesian coordinate in plane

$\hat{x}$  is the unit vector perpendicular to  $x = \text{constant}$  line

$\hat{y}$  is the unit vector perpendicular to  $y = \text{constant}$  line



# Position vector in terms of Cartesian components



- Position **P** can be represented through Cartesian coordinate  $(x, y)$ .
  - From vector addition rule
$$\overrightarrow{OP} = \overrightarrow{OA} + \overrightarrow{AP}$$
- $\overrightarrow{OA} =$  (magnitude of OA) (unit vector along/parallel to X-direction)  $= x\hat{x}$
- $\overrightarrow{AP} =$  (magnitude of AP) (unit vector along/parallel to Y-direction)  $= y\hat{y}$

We may also use the notation:

$$\begin{aligned}\hat{x} &= \hat{i} \\ \hat{y} &= \hat{j}\end{aligned}$$

$$\text{Thus, } \overrightarrow{OP} = \vec{r} = x\hat{x} + y\hat{y}$$



# Velocity and acceleration in Cartesian

Velocity in Cartesian coordinate

$$\vec{v} = \frac{d\vec{r}}{dt} = \frac{d}{dt} (x \hat{x} + y \hat{y})$$

$$\Rightarrow = \dot{x} \hat{x} + x \frac{d\hat{x}}{dt} + \dot{y} \hat{y} + y \frac{d\hat{y}}{dt}$$

$$\Rightarrow \vec{v} = \dot{x} \hat{x} + \dot{y} \hat{y}$$

$$\begin{aligned} \frac{d\hat{x}}{dt} &= 0 \\ \frac{d\hat{y}}{dt} &= 0 \end{aligned}$$

**Standard  
Notations:**

$$\frac{dx}{dt} = \dot{x}$$

Acceleration  $\vec{a} = \frac{d\vec{v}}{dt} = \ddot{x} \hat{x} + \ddot{y} \hat{y}$

Newton's second law in vector form

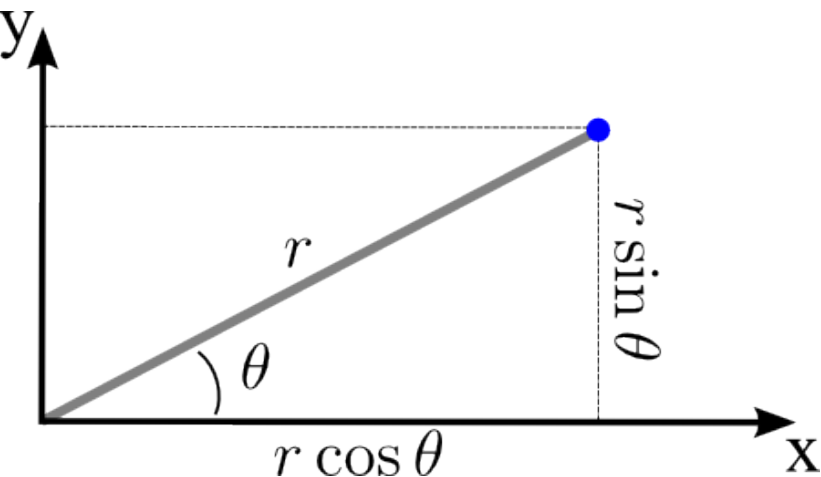
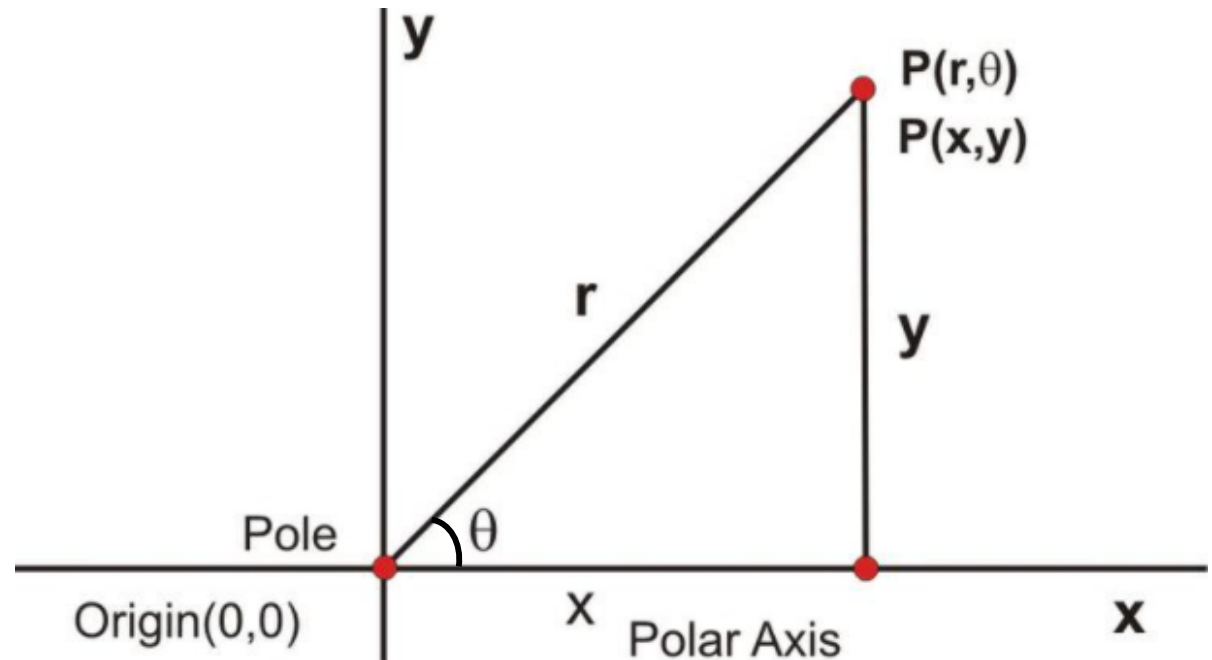
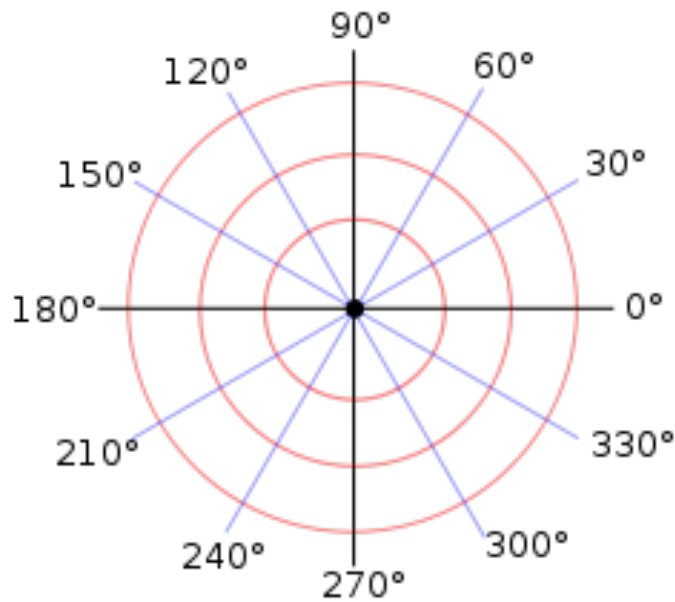
$$\vec{F} = F_x \hat{x} + F_y \hat{y} = m \frac{d\vec{v}}{dt} = m(\ddot{x} \hat{x} + \ddot{y} \hat{y})$$



# Plane Polar Coordinates



# Plane polar coordinates



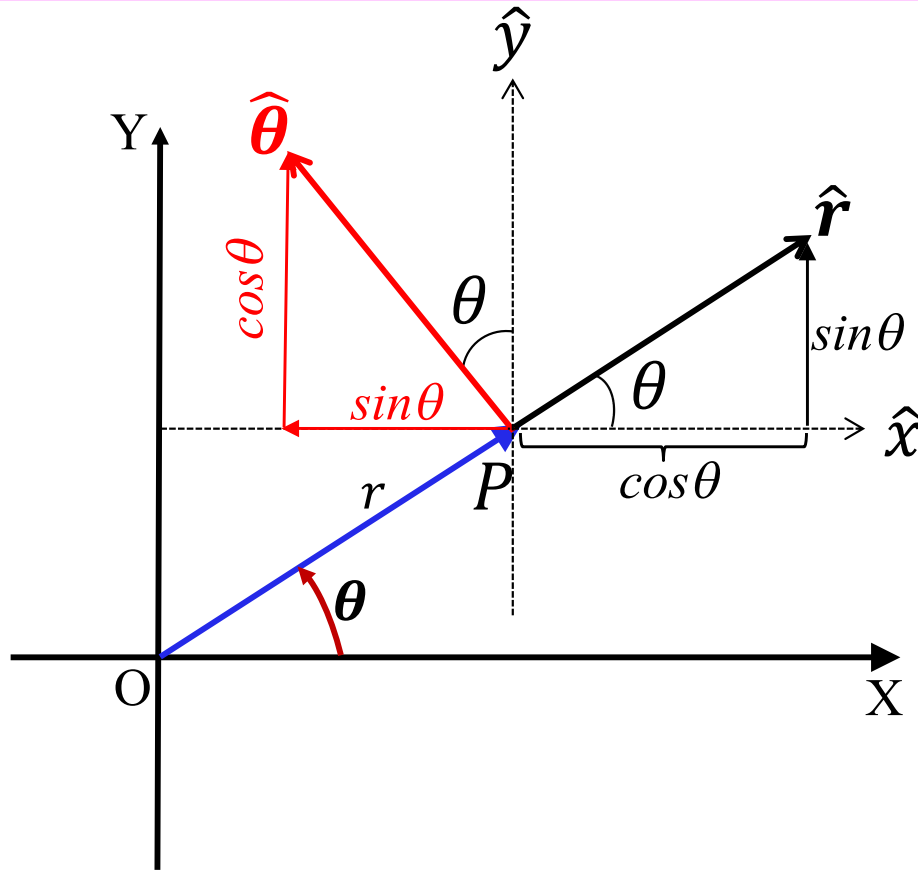
Each point  $P(x, y)$  on the plane can also be represented by its distance,  $r$  from the origin  $O$  and the angle,  $\theta$ ,  $OP$  makes with X-axis.

→ Relationship with Cartesian coordinates

$$x = r \cos \theta \quad \& \quad y = r \sin \theta$$

$$r = (x^2 + y^2)^{1/2} \quad \& \quad \theta = \tan^{-1} \left( \frac{y}{x} \right)$$

# Unit vector in plane polar coordinate



⊙ The unit vectors for plan polar coordinates are  $\hat{r}$  and  $\hat{\theta}$ .

⊙  $\hat{r}$  and  $\hat{\theta}$  are unit vectors along increasing direction of coordinate  $r$  and  $\theta$ .

⊙ The direction of  $\hat{r}$  and  $\hat{\theta}$  varies with position.

$$\hat{r} = \cos \theta \hat{x} + \sin \theta \hat{y}$$

$$\hat{\theta} = -\sin \theta \hat{x} + \cos \theta \hat{y}$$

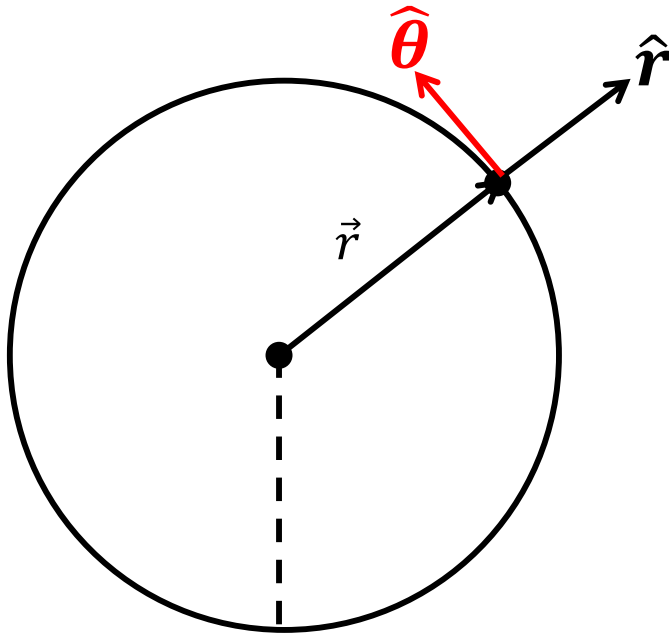
Unit vectors depend only on  $\theta$ .

$\hat{r}$  and  $\hat{\theta}$  are **orthogonal**  $\hat{r} \cdot \hat{\theta} = 0$

Their directions depend on location.



# Unit vector in plane polar coordinate



- ⊙ The position vector  $\vec{r}$  in plane polar coordinates is given by  $\vec{r} = r\hat{r}$ .
- ⊙ In cartesian coordinates  
$$\vec{r} = x\hat{x} + y\hat{y},$$
  
where  $x = r \cos \theta$  &  $y = r \sin \theta$ .
- ⊙ Thus,  $\vec{r} = r \cos \theta \hat{x} + r \sin \theta \hat{y}$

The unit vectors are defined as  $\hat{r} = \left( \frac{\partial \vec{r} / \partial r}{|\partial \vec{r} / \partial r|} \right)$  and  $\hat{\theta} = \left( \frac{\partial \vec{r} / \partial \theta}{|\partial \vec{r} / \partial \theta|} \right)$ .

$$\hat{r} = \cos \theta \hat{x} + \sin \theta \hat{y}$$
$$\hat{\theta} = -\sin \theta \hat{x} + \cos \theta \hat{y}$$

# Unit vectors' relation ( $\hat{r}$ , $\hat{\theta}$ vs $\hat{x}$ , $\hat{y}$ )

$$\begin{aligned}\hat{r} &= \cos \theta \hat{x} + \sin \theta \hat{y} \\ \hat{\theta} &= -\sin \theta \hat{x} + \cos \theta \hat{y}\end{aligned}$$

Multiply Eq. for  $\hat{r}$  with  $\cos \theta \rightarrow \cos \theta \hat{r} = \cos^2 \theta \hat{x} + \cos \theta \sin \theta \hat{y}$

Multiply Eq. for  $\hat{\theta}$  with  $\sin \theta \rightarrow \sin \theta \hat{\theta} = -\sin^2 \theta \hat{x} + \sin \theta \cos \theta \hat{y}$

Subtracting  $\rightarrow$

$$\hat{x} = \cos \theta \hat{r} - \sin \theta \hat{\theta}$$

Multiply Eq. for  $\hat{r}$  with  $\sin \theta \rightarrow \sin \theta \hat{r} = \cos \theta \sin \theta \hat{x} + \sin^2 \theta \hat{y}$

Multiply Eq. for  $\hat{\theta}$  with  $\cos \theta \rightarrow \cos \theta \hat{\theta} = -\sin \theta \cos \theta \hat{x} + \cos^2 \theta \hat{y}$

Adding  $\rightarrow$

$$\hat{y} = \sin \theta \hat{r} + \cos \theta \hat{\theta}$$

$$\hat{x} = \cos \theta \hat{r} - \sin \theta \hat{\theta}$$

$$\hat{y} = \sin \theta \hat{r} + \cos \theta \hat{\theta}$$



# Derivative of Unit vectors in plane polar

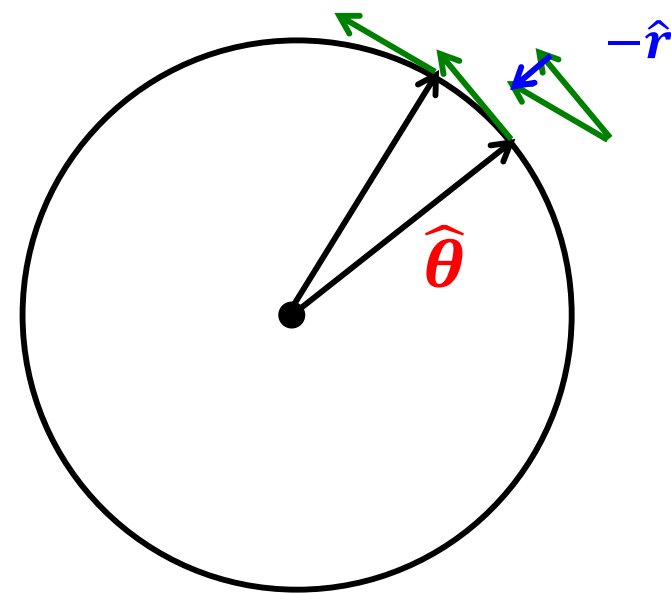
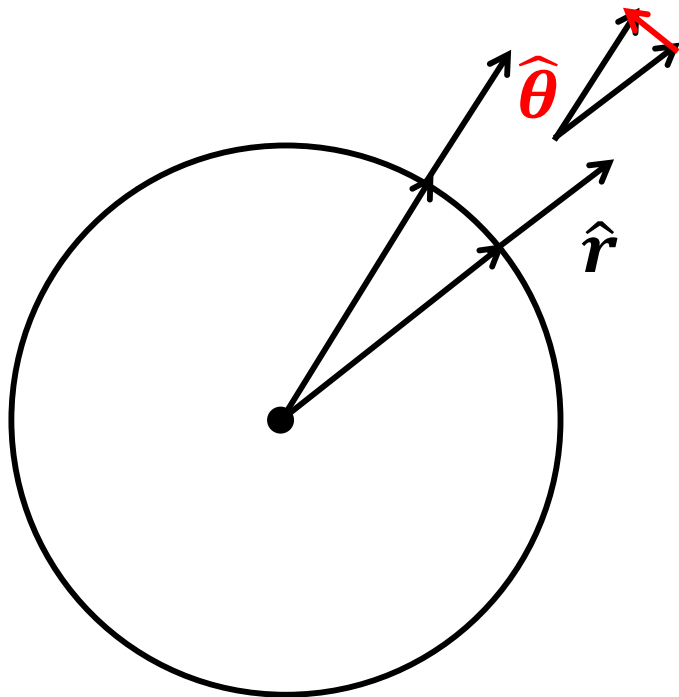
The unit vectors are function of  $\theta$  only.

$$\hat{r} = \cos \theta \hat{x} + \sin \theta \hat{y}$$

$$\hat{\theta} = -\sin \theta \hat{x} + \cos \theta \hat{y}$$

$$\frac{\partial \hat{r}}{\partial \theta} = -\sin \theta \hat{x} + \cos \theta \hat{y} = \hat{\theta}$$

$$\frac{\partial \hat{\theta}}{\partial \theta} = -\cos \theta \hat{x} - \sin \theta \hat{y} = -(\cos \theta \hat{x} + \sin \theta \hat{y}) = -\hat{r}$$



# Velocity in plane polar coordinate

Velocity  $\vec{v} = \frac{d\vec{r}}{dt}$   $\vec{v} = \frac{d\vec{r}}{dt} = \frac{d(r\hat{r})}{dt}$

**Standard  
Notations:**

$$\frac{dr}{dt} = \dot{r}$$

$$\frac{d^2\theta}{dt^2} = \ddot{\theta}$$

$$= \frac{dr}{dt} \hat{r} + r \frac{d\hat{r}}{dt}$$

$$= \dot{r} \hat{r} + r \frac{\partial \hat{r}}{\partial \theta} \frac{d\theta}{dt}$$

$$\boxed{\vec{v} = \dot{r} \hat{r} + r \dot{\theta} \hat{\theta}}$$

Since,

$$\frac{\partial \hat{r}}{\partial \theta} = \hat{\theta}$$

$$\frac{\partial \hat{\theta}}{\partial \theta} = -\hat{r}$$

Radial component,  $\dot{r}$  and tangential/transverse component,  $r\dot{\theta}$



# Acceleration in plane polar coordinate

$$\vec{a} = \frac{d\vec{v}}{dt}$$

Note:

$$\frac{\partial \hat{r}}{\partial \theta} = \hat{\theta} \quad \& \quad \frac{\partial \hat{\theta}}{\partial \theta} = -\hat{r}$$

$$\begin{aligned} \frac{d\vec{v}}{dt} &= \frac{d}{dt} (\dot{r}\hat{r} + r\dot{\theta}\hat{\theta}) = \frac{dr}{dt}\hat{r} + \dot{r}\frac{d\hat{r}}{dt} + \frac{dr}{dt}\dot{\theta}\hat{\theta} + r\ddot{\theta}\hat{\theta} + r\dot{\theta}\frac{d\hat{\theta}}{dt} \\ &= \frac{dr}{dt}\hat{r} + \dot{r}\frac{d\hat{r}}{dt} + \dot{r}\dot{\theta}\hat{\theta} + r\ddot{\theta}\hat{\theta} + r\dot{\theta}\frac{d\hat{\theta}}{dt} \\ &= \dot{r}\hat{r} + \dot{r}\frac{\partial \hat{r}}{\partial \theta}\frac{d\theta}{dt} + \dot{r}\dot{\theta}\hat{\theta} + r\ddot{\theta}\hat{\theta} + r\dot{\theta}\frac{\partial \hat{\theta}}{\partial \theta}\frac{d\theta}{dt} \\ &= \dot{r}\hat{r} + \dot{r}\dot{\theta}\hat{\theta} + \dot{r}\dot{\theta}\hat{\theta} + r\ddot{\theta}\hat{\theta} - r\dot{\theta}^2\hat{r} \end{aligned}$$

$$\vec{a} = (\ddot{r} - r\dot{\theta}^2)\hat{r} + (2\dot{r}\dot{\theta} + r\ddot{\theta})\hat{\theta}$$



# Acceleration in plane polar coordinate

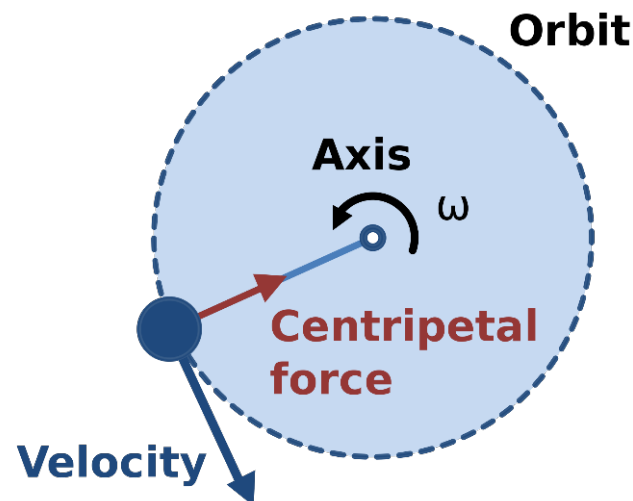
$$\vec{a} = (\ddot{r} - r\dot{\theta}^2)\hat{r} + (2\dot{r}\dot{\theta} + r\ddot{\theta})\hat{\theta}$$

Radial component  $\ddot{r} - r\dot{\theta}^2$

( $-r\dot{\theta}^2$  is the familiar *Centripetal contribution*!)

Tangential component:  $2\dot{r}\dot{\theta} + r\ddot{\theta}$   $r\ddot{\theta}$  Intuitively clear

$2\dot{r}\dot{\theta}$  is called the *Coriolis contribution*!



# Newton's law in plane polar coordinate

$$\vec{F} = m \frac{d\vec{v}}{dt}$$

$$\vec{F} = F_r \hat{r} + F_\theta \hat{\theta} = m[(\ddot{r} - r\dot{\theta}^2)\hat{r} + (2\dot{r}\dot{\theta} + r\ddot{\theta})\hat{\theta}]$$

Newton's law for **radial** direction

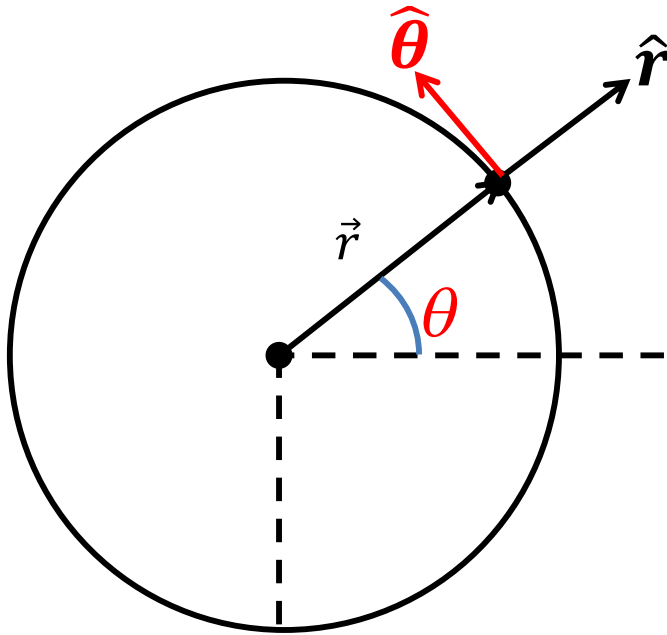
$$F_r = m(\ddot{r} - r\dot{\theta}^2)$$

Newton's law for **tangential** direction

$$F_\theta = m(2\dot{r}\dot{\theta} + r\ddot{\theta})$$



# Choice of the coordinate system??



Motion in circular trajectory

Equation of trajectory in polar coordinate

$$r = R = \text{constant}; \theta = \omega t + \frac{1}{2}\alpha t^2$$

The velocity components are

$$v_r = \dot{r} = 0; v_\theta = r\dot{\theta} = R(\omega + \alpha t) = v$$

Acceleration components are

$$a_r = \ddot{r} - r\dot{\theta}^2 = -R(\omega + \alpha t)^2$$

$$a_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta} = R\alpha = a_t$$

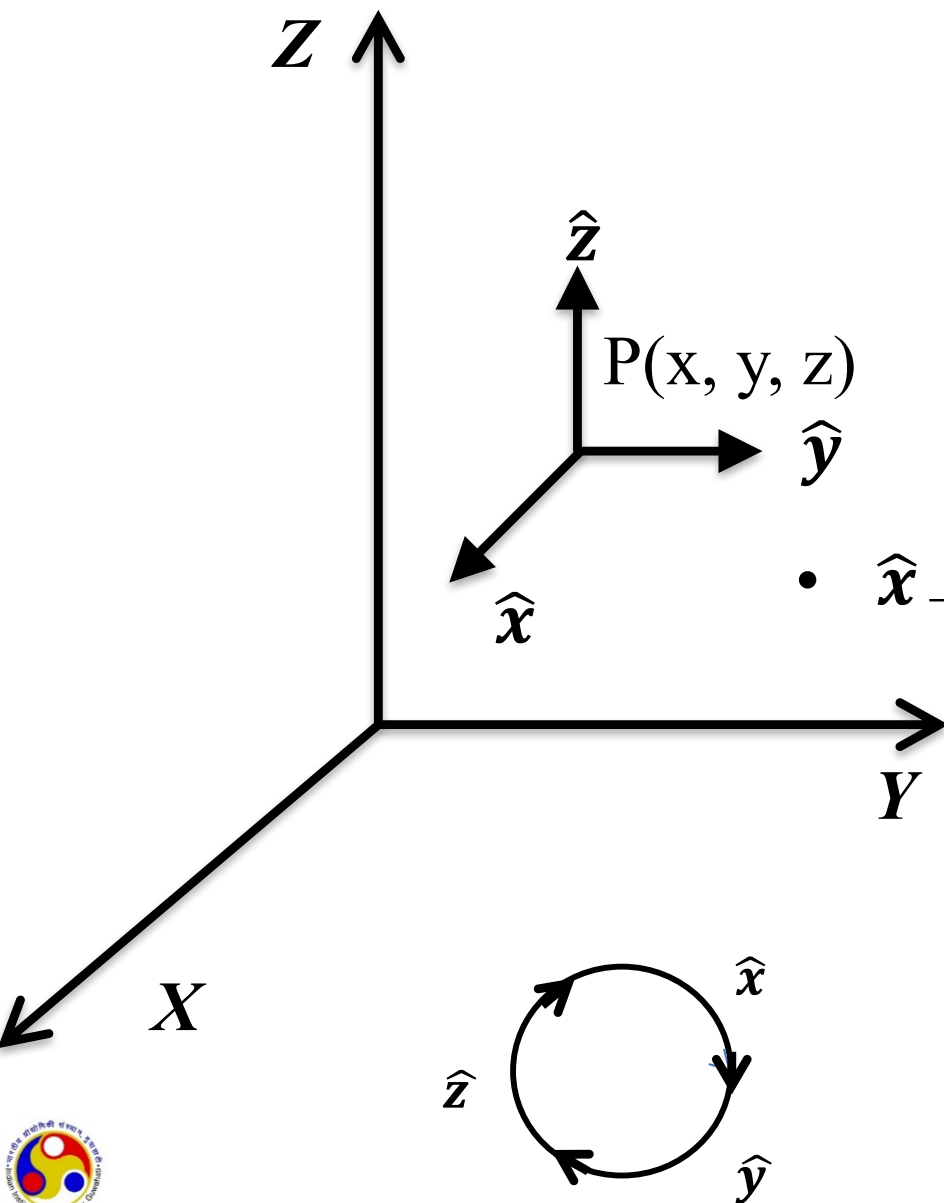
Equation of trajectory in Cartesian coordinate

$$x = R \cos\left(\omega t + \frac{1}{2}\alpha t^2\right); y = R \sin\left(\omega t + \frac{1}{2}\alpha t^2\right);$$

Velocity components are

$$v_x = -R(\omega + \alpha t) \sin\left(\omega t + \frac{1}{2}\alpha t^2\right); v_y = R(\omega + \alpha t) \cos\left(\omega t + \frac{1}{2}\alpha t^2\right)$$

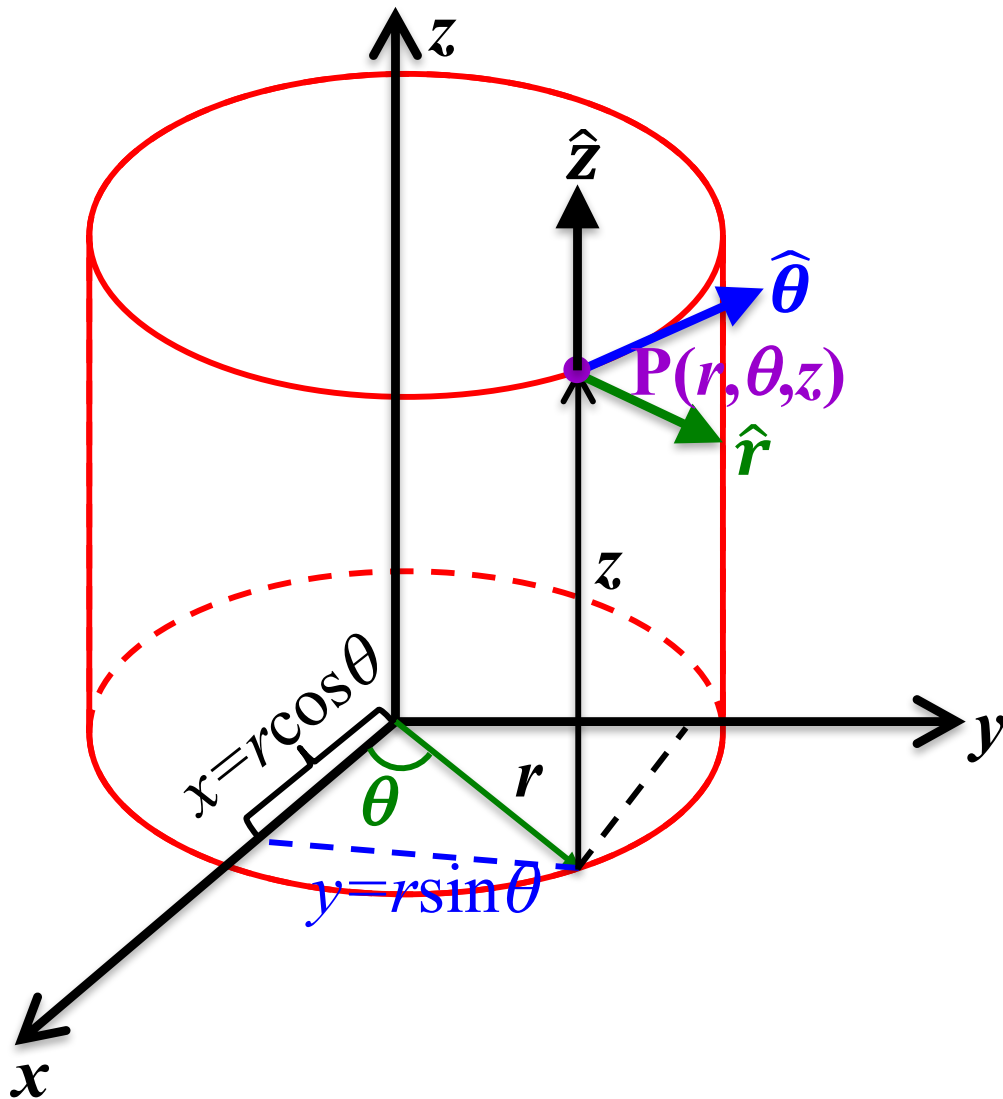
# Cartesian co-ordinate system in 3D



- In Cartesian coordinate system, one can define three unit vectors  $\hat{x}$ ,  $\hat{y}$  and  $\hat{z}$  at every point.
- $\hat{x}$ ,  $\hat{y}$  and  $\hat{z}$  are unit vectors in the increasing direction of  $x$ ,  $y$  and  $z$ .
- $\hat{x} \perp$  to the  $YZ$  plane  $\hat{y} \perp$  to the  $XZ$  plane and  $\hat{z} \perp$  to the  $XY$  plane
- $\hat{x}$ ,  $\hat{y}$  and  $\hat{z}$  are orthogonal, and directed in the same direction at every points.

$$\hat{x} \times \hat{y} = \hat{z}; \hat{y} \times \hat{z} = \hat{x}; \hat{z} \times \hat{x} = \hat{y};$$

# Cylindrical Coordinate System



**How to locate a point 'P' in space ?**

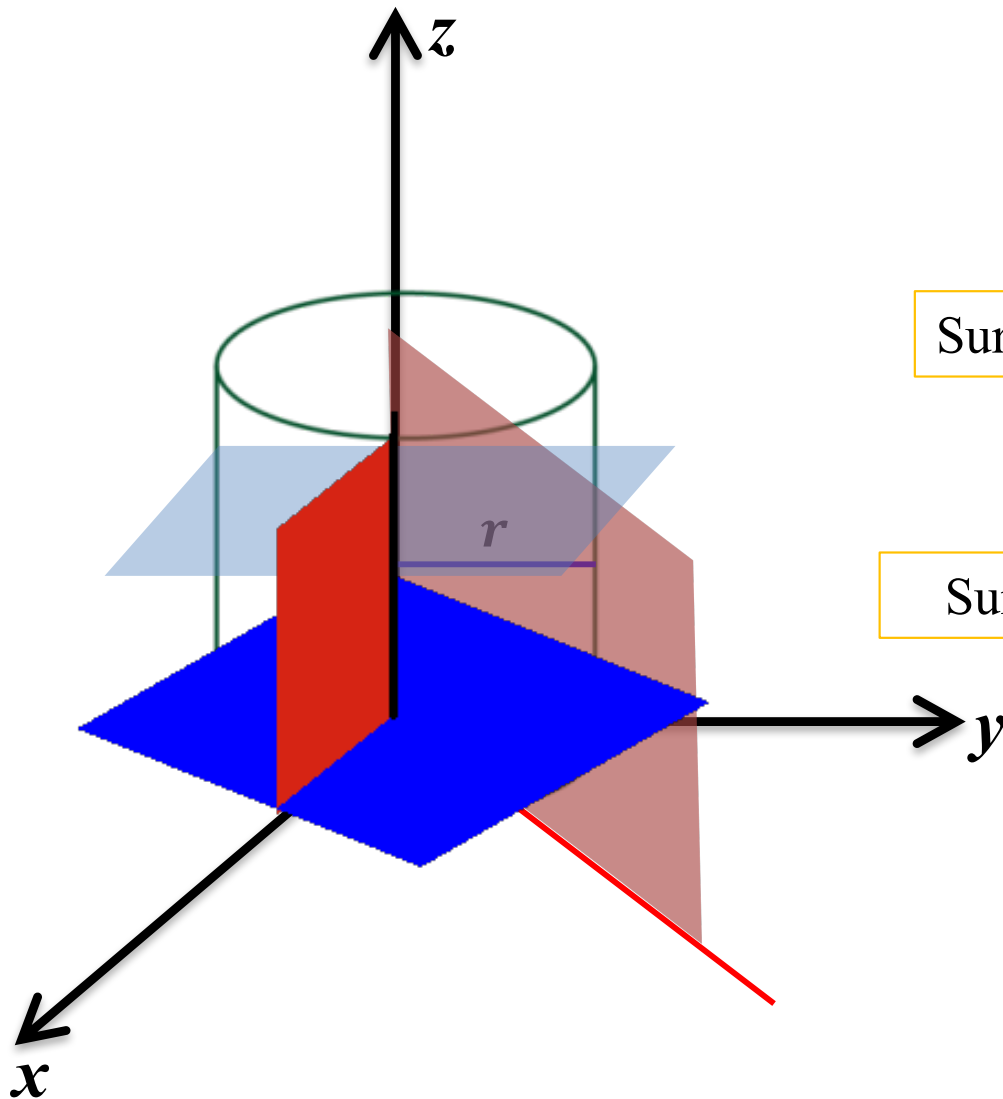
$(r, \theta, z)$  coordinates system is known as cylindrical coordinate system

$(r, \theta)$  Coordinate of the foot of the point in XY plane

**z-Height from the XY plane**

$\hat{r}$ ,  $\hat{\theta}$ , and  $\hat{z}$  are unit vectors along increasing direction of  $r$ ,  $\theta$ , and  $z$ .

# Cylindrical Coordinate System



Note that point  $(r, \theta, z)$  is at the intersection of three surfaces:

The surface with  $r = \text{Constant}$ ; Cylinder

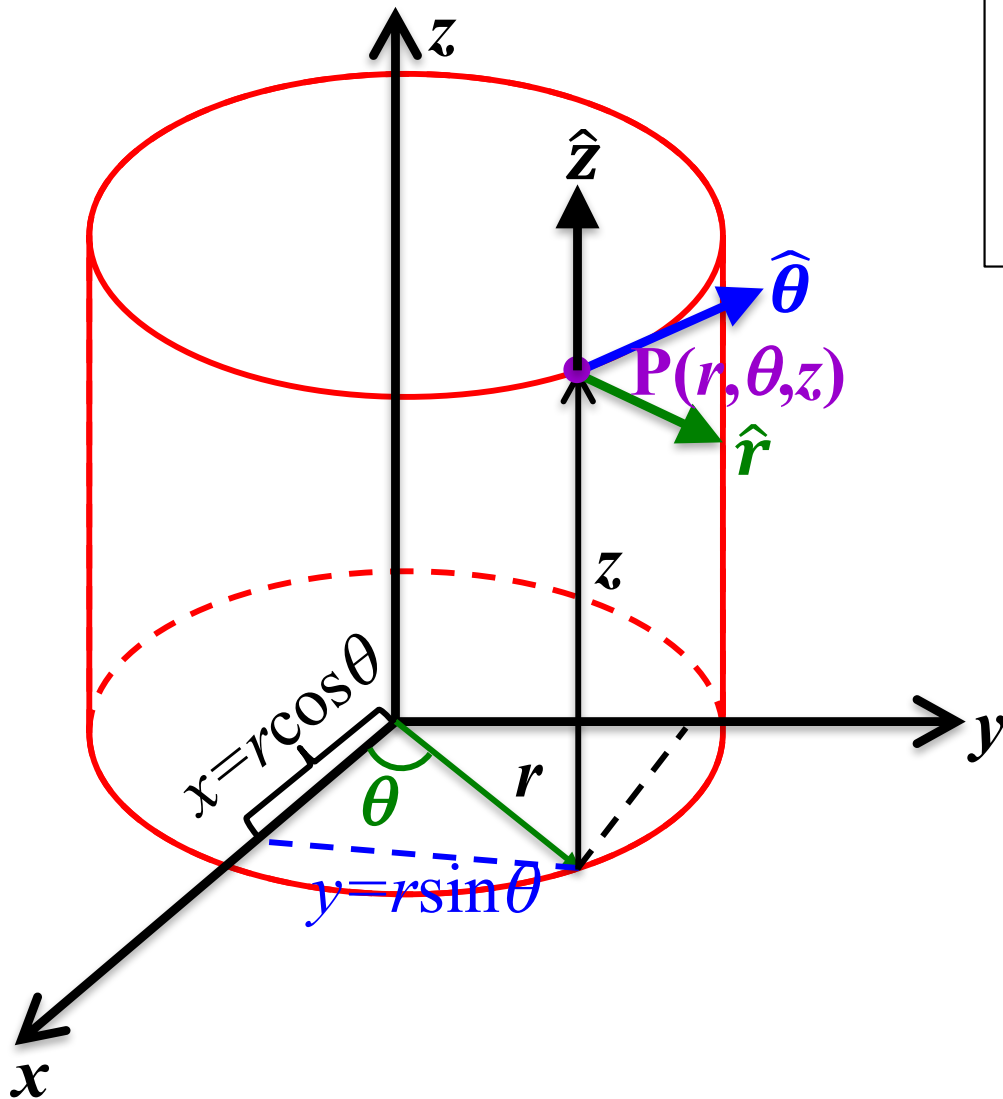
Surface with  $\theta = \text{Constant}$ ; Plane containing Z-axis

Surface  $z = \text{Constant}$ . Plane Perpendicular Z-axis

The three orthogonal components,  $r$  (green),  $\theta$  (red), and  $z$  (blue), each increasing at a constant rate. The point is at the intersection between the three colored surfaces.



# Cylindrical Coordinate System



Transformation equation is very similar to polar coordinate with additional z-coordinate.

$$\begin{aligned}x &= r \cos \theta \\y &= r \sin \theta \\z &= z\end{aligned}$$

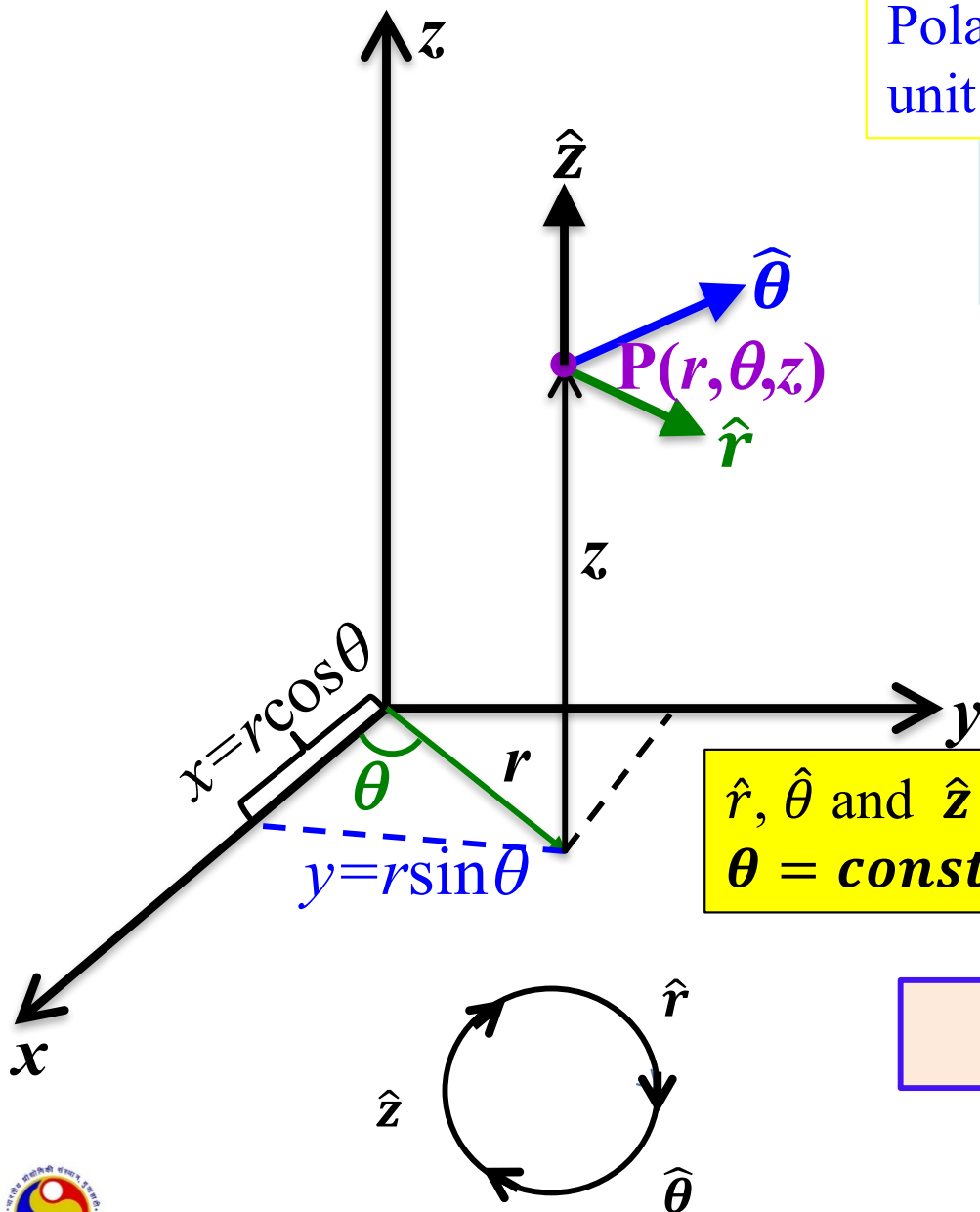
**Reverse transformation**

$$\begin{aligned}r &= (x^2 + y^2)^{1/2} \\ \theta &= \tan^{-1} \frac{y}{x} \\ z &= z\end{aligned}$$

**Note:** Instead of  $(r, \theta)$ , many books use notation  $(\rho, \varphi)$ .



# Cylindrical Coordinate System



Polar coordinate unit vectors  $(\hat{r}, \hat{\theta})$  + additional unit vector in the  $z$  -direction.

$\hat{r}, \hat{\theta}$  and  $\hat{z}$  are unit vectors along increasing direction of coordinates  $r, \theta$  and  $z$ .

$$\begin{aligned}\hat{r} &= \cos \theta \hat{x} + \sin \theta \hat{y} \\ \hat{\theta} &= -\sin \theta \hat{x} + \cos \theta \hat{y} \\ \hat{z} &= \hat{z}\end{aligned}$$

$\hat{r}, \hat{\theta}$ , and  $\hat{z}$  are **orthogonal**, the directions of  $\hat{r}, \hat{\theta}$ , depend on location but  $\hat{z}$  is fixed

$\hat{r}, \hat{\theta}$  and  $\hat{z}$  are **perpendicular** to surfaces  $r = \text{constant}$ ;  $\theta = \text{constant}$  and  $z = \text{constant}$ , respectively.

$$\hat{r} \times \hat{\theta} = \hat{z}; \quad \hat{\theta} \times \hat{z} = \hat{r}; \quad \hat{z} \times \hat{r} = \hat{\theta}$$



# Position, Velocity, Acceleration, Newton's law in cylindrical coordinate system

Vector components are very similar to polar coordinate + z component

Position vector



$$\vec{r} = r\hat{r} + z\hat{z}$$

Velocity



$$\vec{v} = \dot{r}\hat{r} + r\dot{\theta}\hat{\theta} + \dot{z}\hat{z}$$

$$\frac{d\hat{z}}{dt} = 0$$

Acceleration



$$\vec{a} = (\ddot{r} - r\dot{\theta}^2)\hat{r} + (2\dot{r}\dot{\theta} + r\ddot{\theta})\hat{\theta} + \ddot{z}\hat{z}$$

Newton's law



$$\begin{aligned}\vec{F} &= F_r\hat{r} + F_\theta\hat{\theta} + F_z\hat{z} \\ &= m[(\ddot{r} - r\dot{\theta}^2)\hat{r} + (2\dot{r}\dot{\theta} + r\ddot{\theta})\hat{\theta} + \ddot{z}\hat{z}]\end{aligned}$$



